

DEEP LEARNING LAB

EXPERIMENT 3

Probability Distributions and Bayes' Theorem

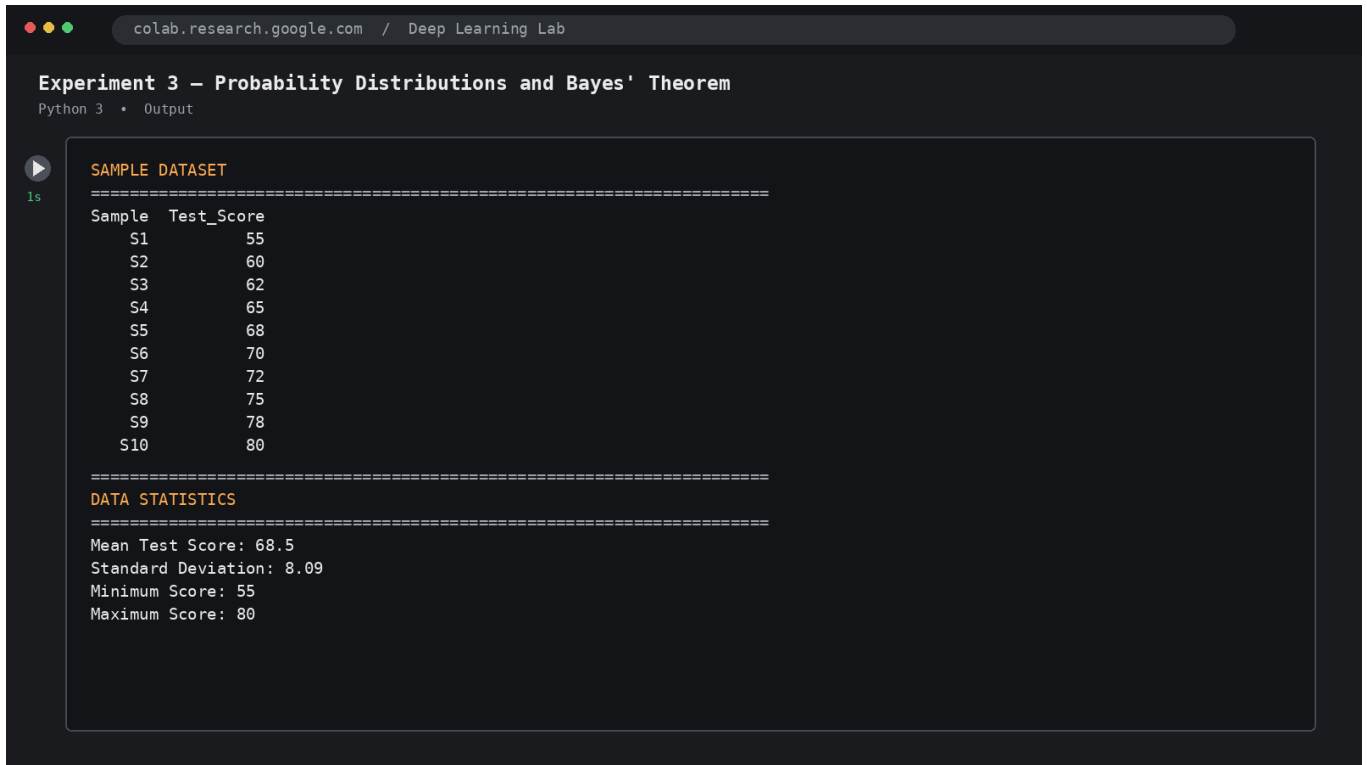


Figure 1: Sample Dataset and Data Statistics

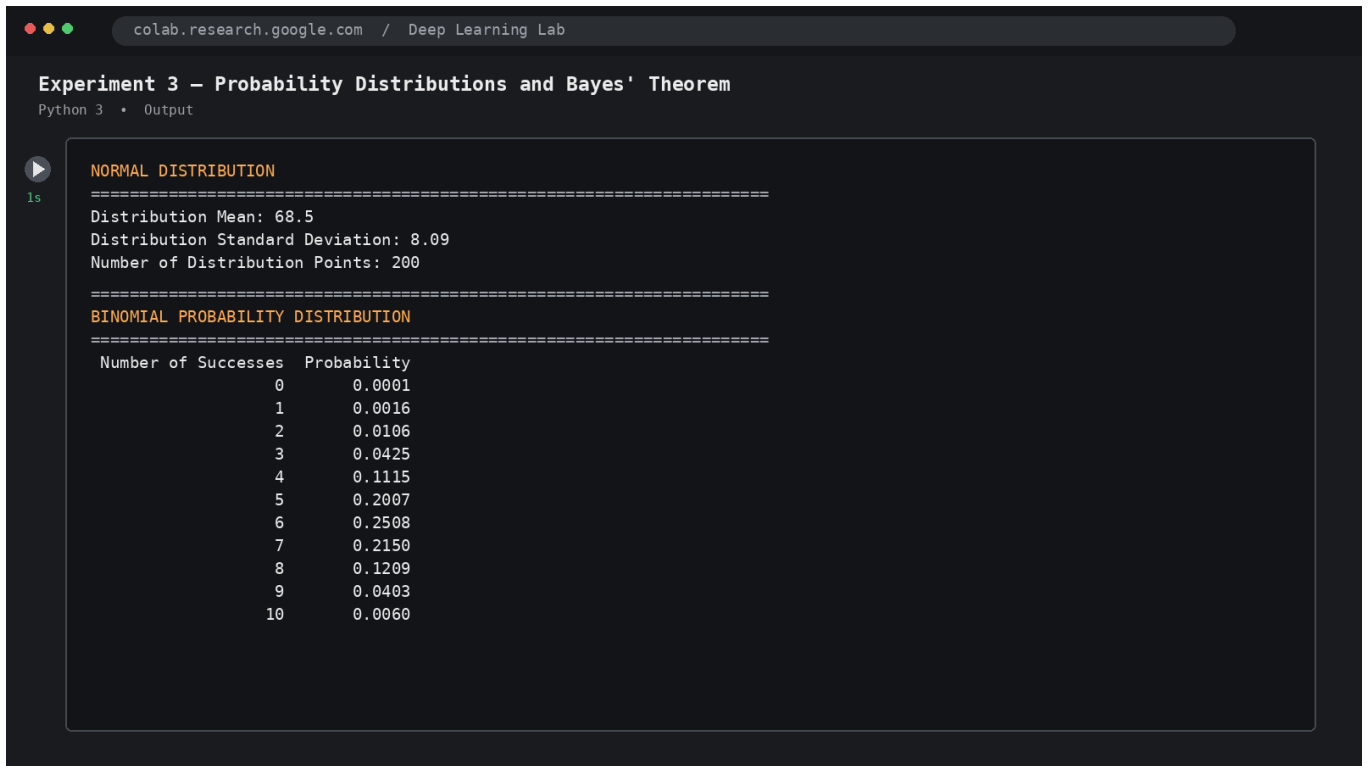


Figure 2: Normal Distribution and Binomial Probability Distribution

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Experiment 3 – Probability Distributions and Bayes' Theorem

Python 3 • Output

▶

1s

BAYES' THEOREM

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P(A) - Prior Probability: 0.01
P(B|A) - True Positive Probability: 0.95
P(B|not A) - False Positive Probability: 0.05
P(B) - Probability of Positive Test: 0.059
P(A|B) - Posterior Probability: 0.161

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BAYES THEOREM EVALUATION TABLE

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Probability	Value
P(A)	0.010
P(B A)	0.950
P(B not A)	0.050
P(B)	0.059
P(A B)	0.161

Figure 3: Bayes' Theorem and Evaluation Table

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PROBABILITY STATISTICS
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Sum of Binomial Probabilities: 1.0
Expected Number of Successes: 6.0
Bayes Posterior Probability (%): 16.1

FINAL RESULT
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Probability distributions were successfully generated
and analyzed using NumPy and SciPy.

The experiment demonstrated:
1. Sample dataset preparation
2. Mean and standard deviation calculation
3. Normal probability distribution
4. Binomial probability distribution
5. Bayes' theorem calculation
6. Prior probability
7. Posterior probability
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Figure 4: Probability Statistics and Final Result

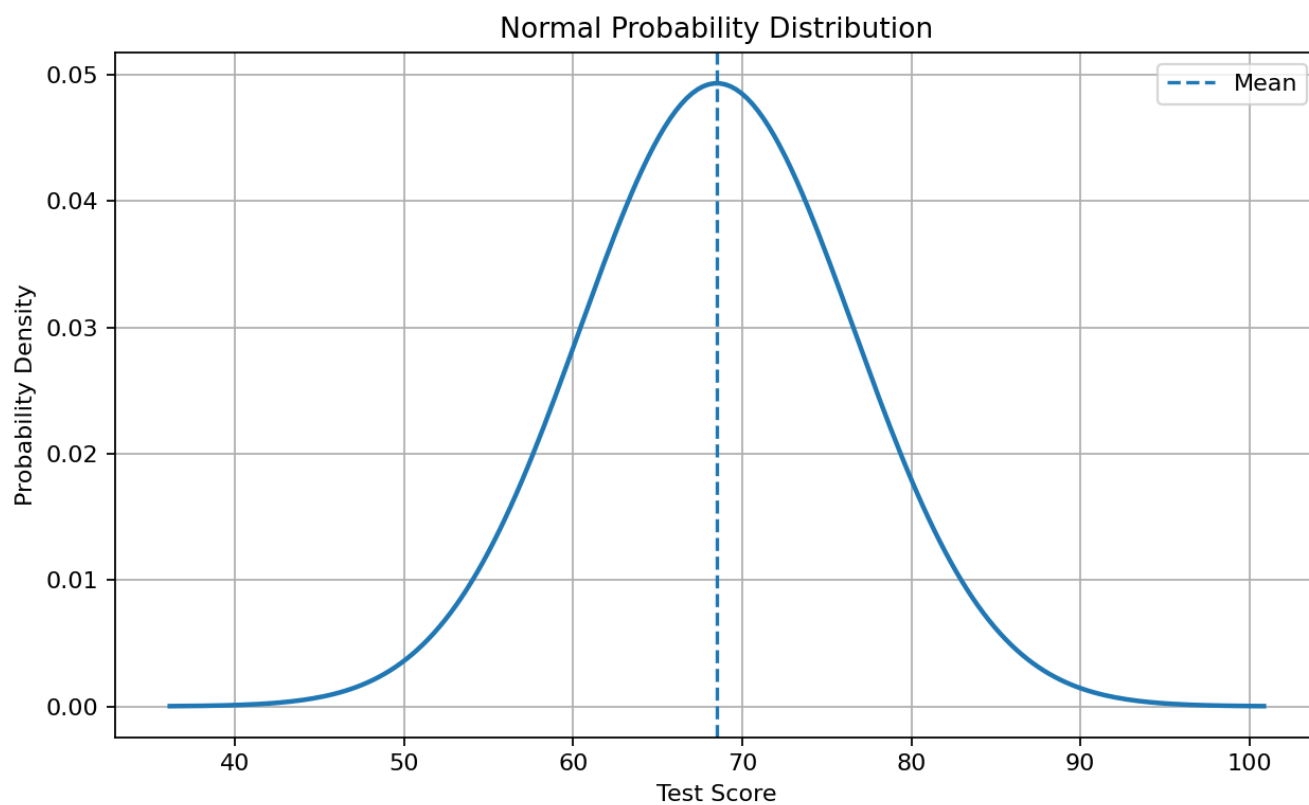


Figure 5: Normal Probability Distribution

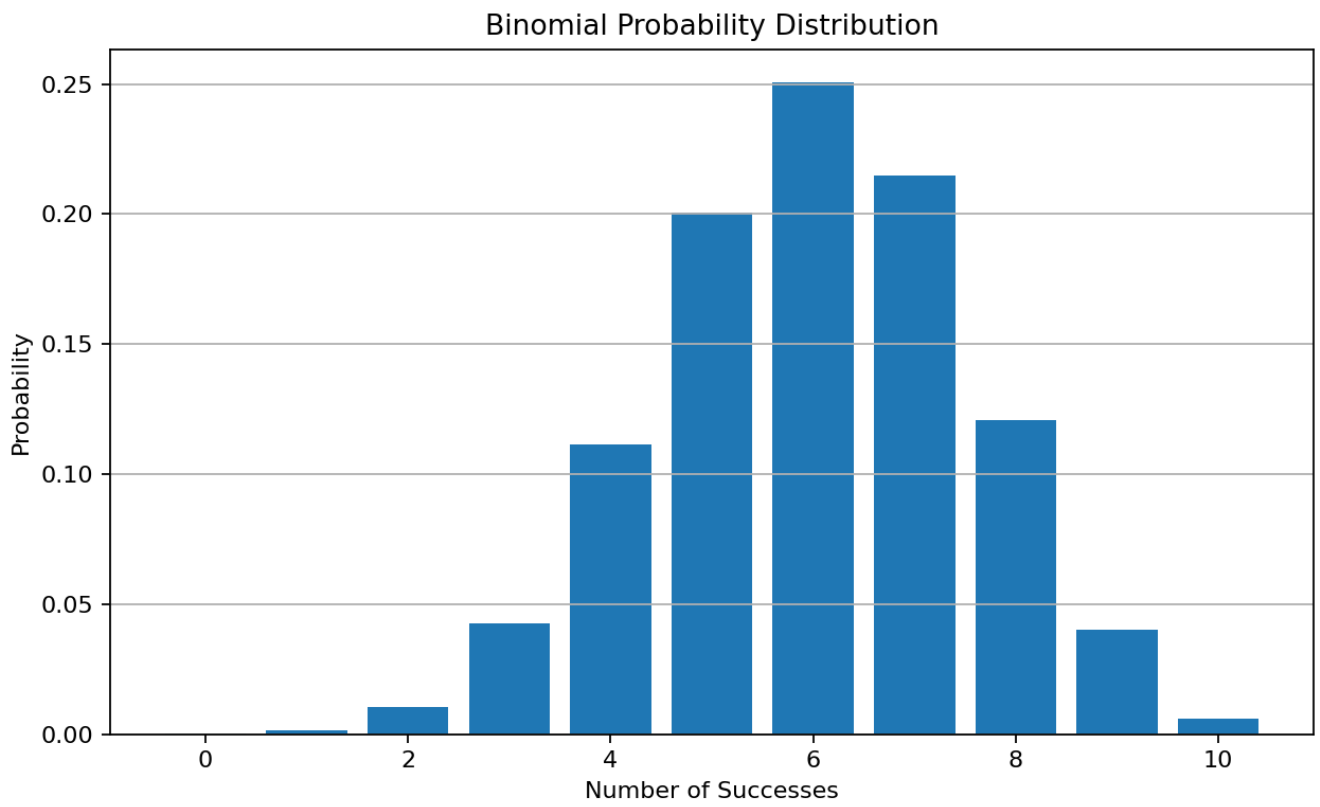


Figure 6: Binomial Probability Distribution

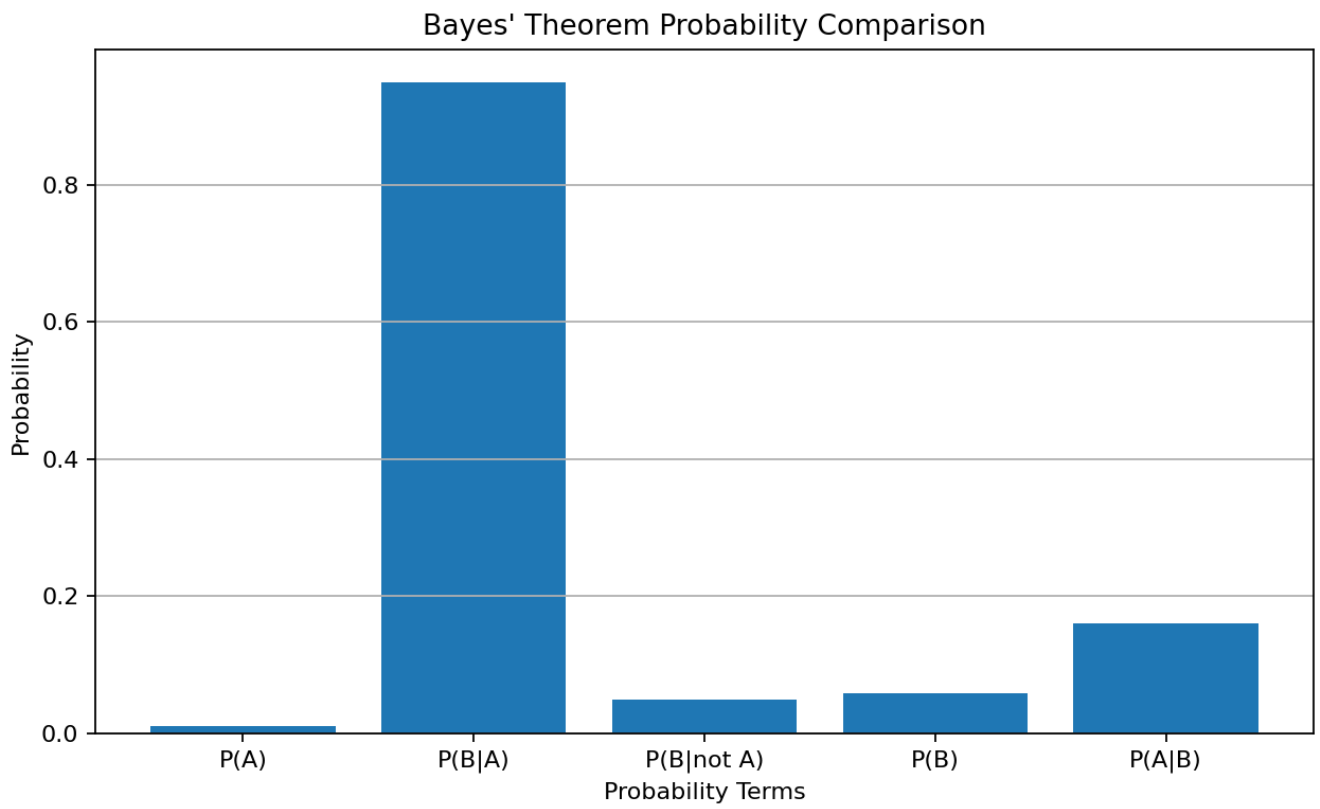


Figure 7: Bayes' Theorem Probability Comparison

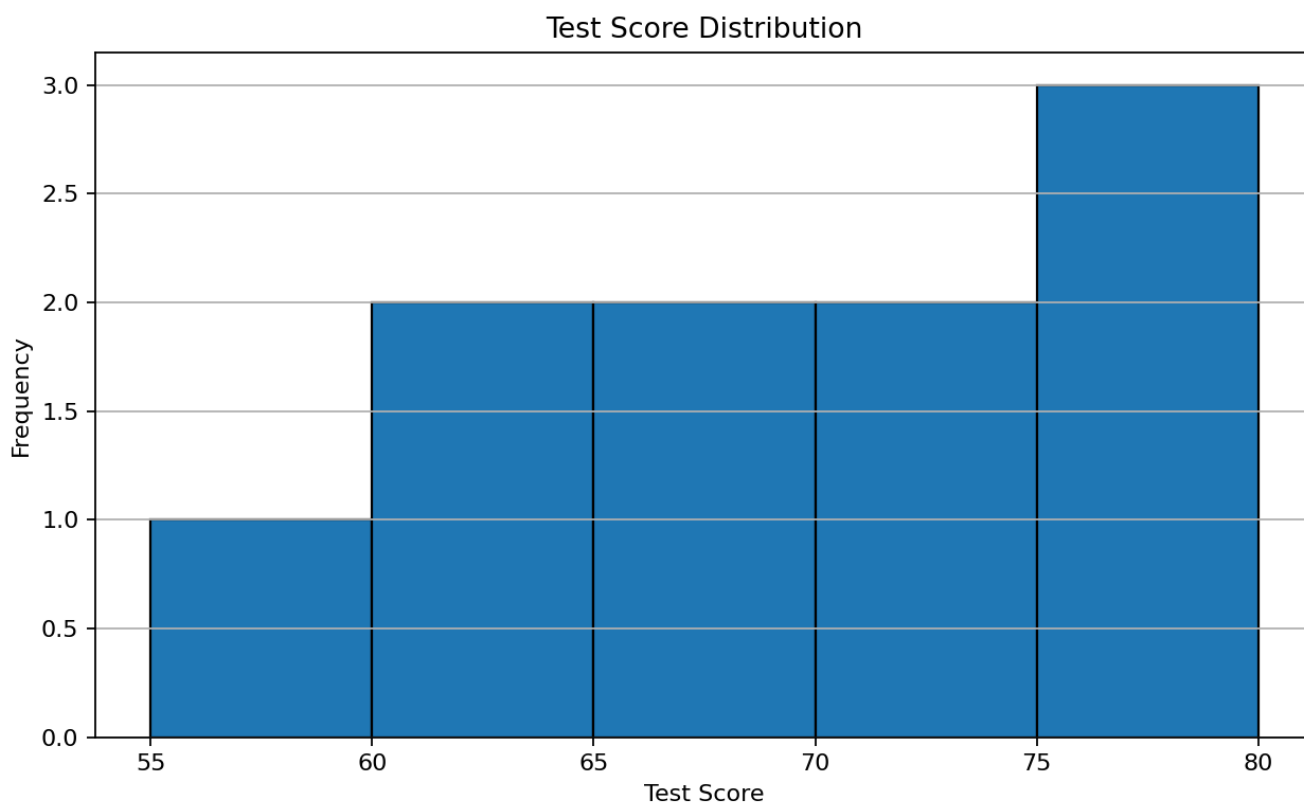


Figure 8: Test Score Distribution